

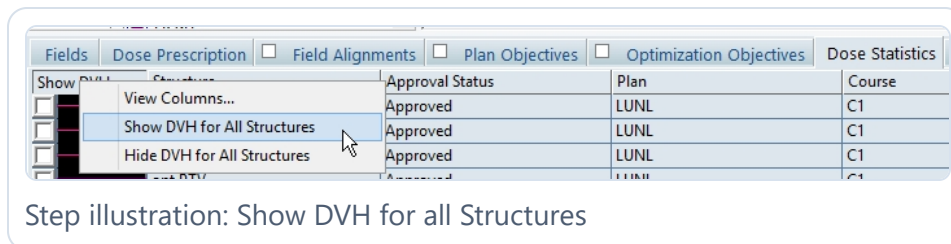
Auto Fill SABR Worksheet Version II

Version date: 2026-05-11

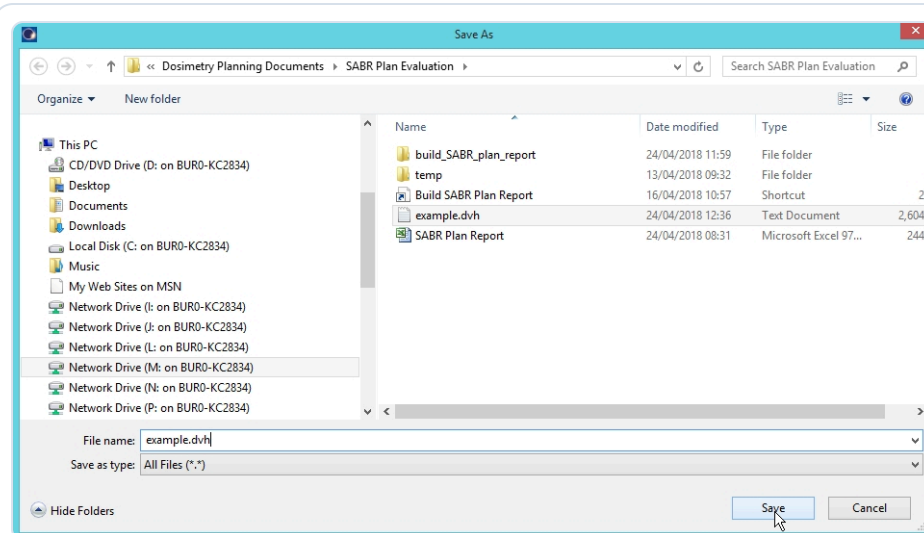
PlanEvaluation reads an exported DVH file, matches plan elements to a selected SABR report definition, and writes values into an Excel worksheet template for review and documentation.

1) Save the DVH Data

1. Open the DVH view in External Beam Planning.
2. In the "Dose Statistics" Tab Right-Click and select "Show DVH for all Structures".



3. DVH values will be calculated.
4. Right-Click within the DVH and select "Absolute Dose".
5. Right-Click within the DVH and select "Export DVH in Tabular Format...".
6. A "Save As" dialogue box will appear.
7. Select the folder "M:\Dosimetry Planning Documents\SABR Plan Evaluation\DVH Files".
8. Set "Save as type" to "All Files".
9. Save with a file name ending in ".dvh".



Step illustration: Save As dialog with All Files and .dvh filename

Confirm the selected DVH export is the current approved plan and fractionation before running the report.

2) Run the Plan Evaluation program

Program name

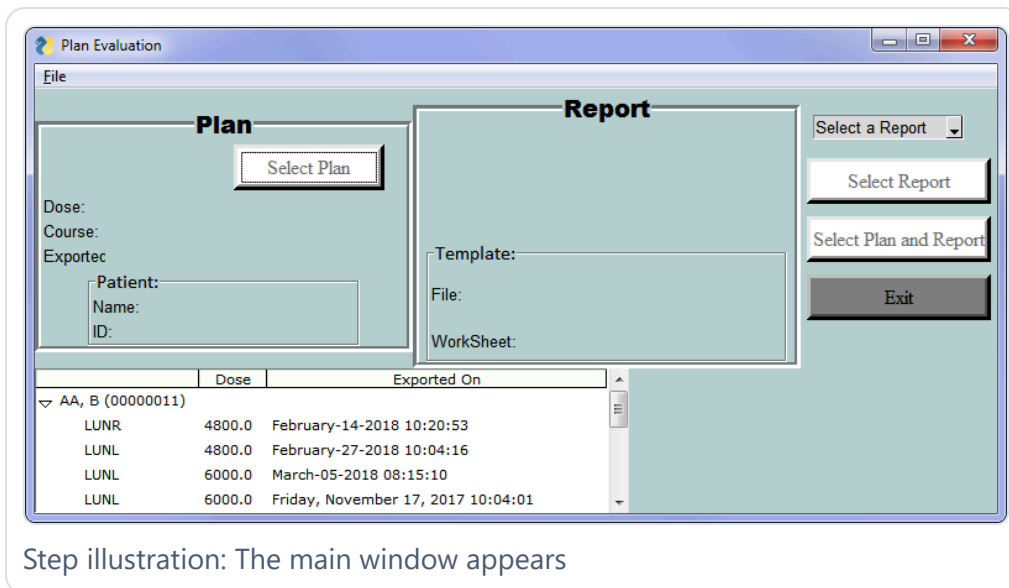
- PlanEvaluation



Start method

- Run PlanEvaluation.bat in the package root.
- Run PlanEvaluation.py from the same folder with the configured Python environment.

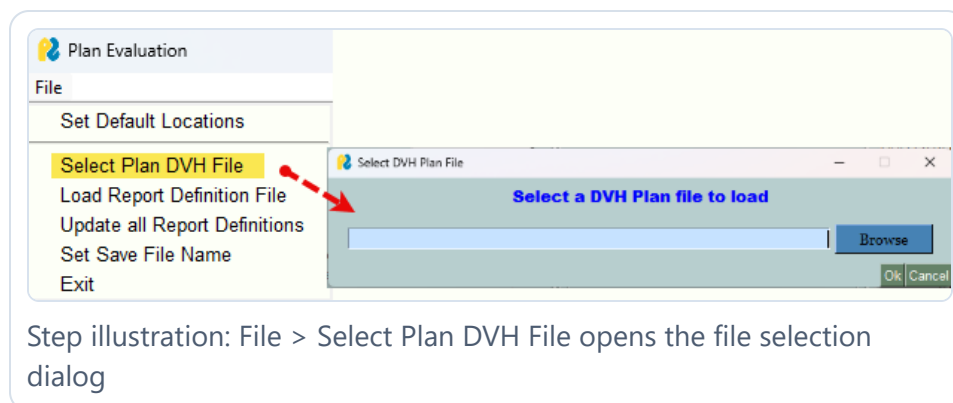
On startup, the app loads configuration from `PlanEvaluationConfig.xml`, report definitions from `Data/Reports.pkl`, and available DVH plans from the configured DVH folder.



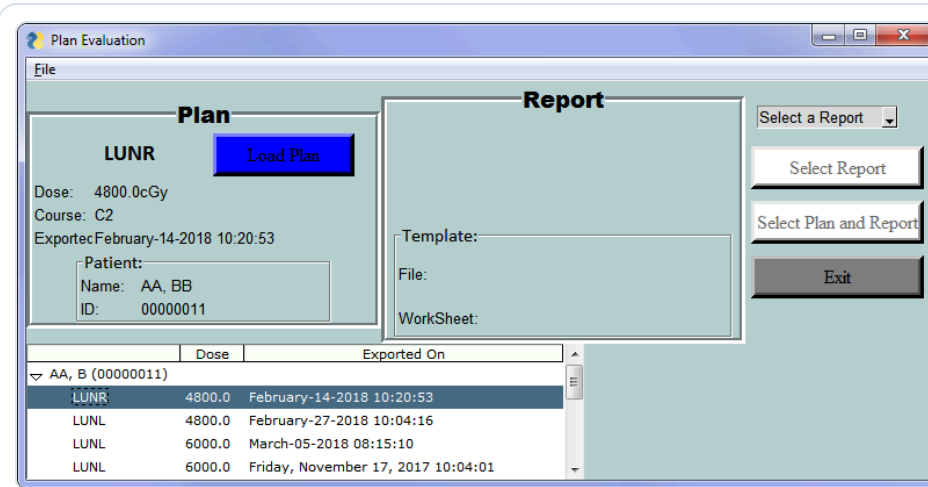
3) Main Window

After starting "PlanEvaluation", the main window appears.

1. Select the dvh file from the list in the bottom left of the window. (This list reflects dvh files in the configured DVH directory.)
2. To select a DVH file from a different folder, use **File > Select Plan DVH File**. A dialog will appear — type or browse to the file path and click **Ok**.

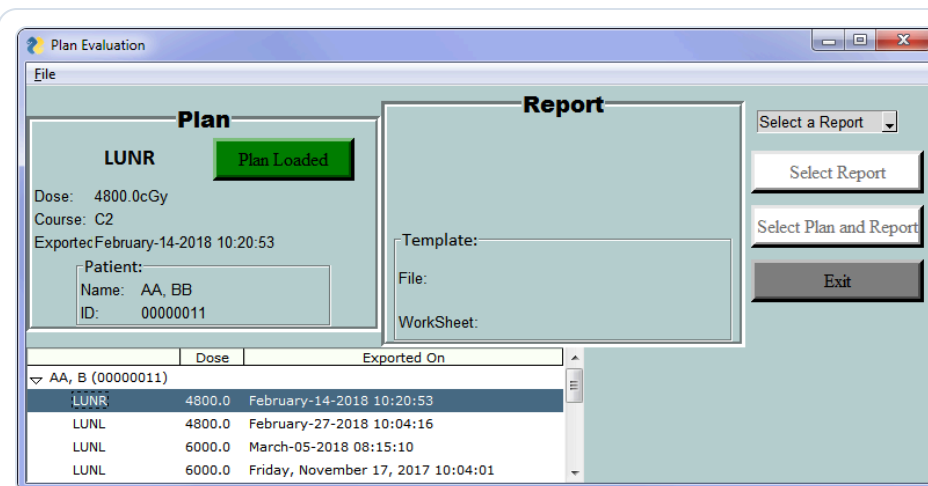


3. Click Load Plan to import the dvh file data.



Step illustration: Select plan and click Load Plan

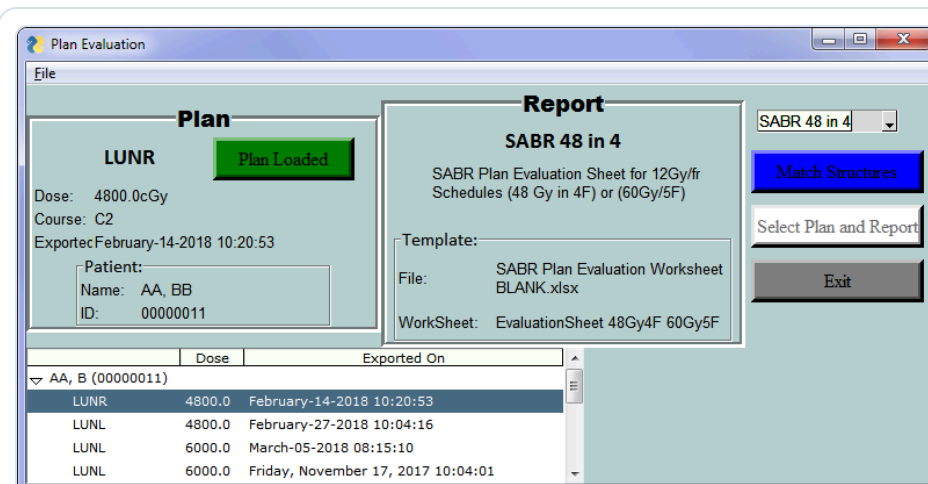
4. The button label should change to Plan Loaded.



Step illustration: Plan Loaded

5. Select a report from the report drop-down list.

6. Click Match Structures to run automatic matching and review manual matching.

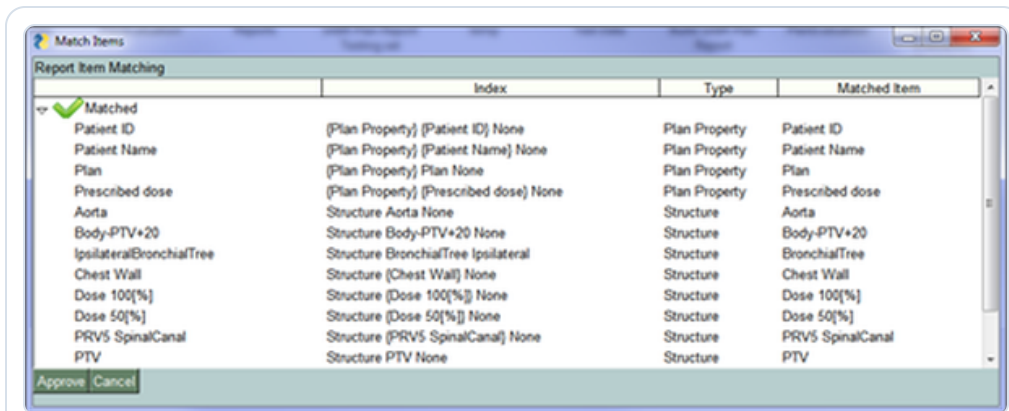


Step illustration: Select report and click Match Structures

4) Matching Structures And Values

In the manual matching window, unmatched items can be manually mapped, directly entered as values, or cleared.

After clicking **Match Structures**, a new window will pop up allowing you to match the plan report items with structure names.



Step illustration: Match window with grouped items

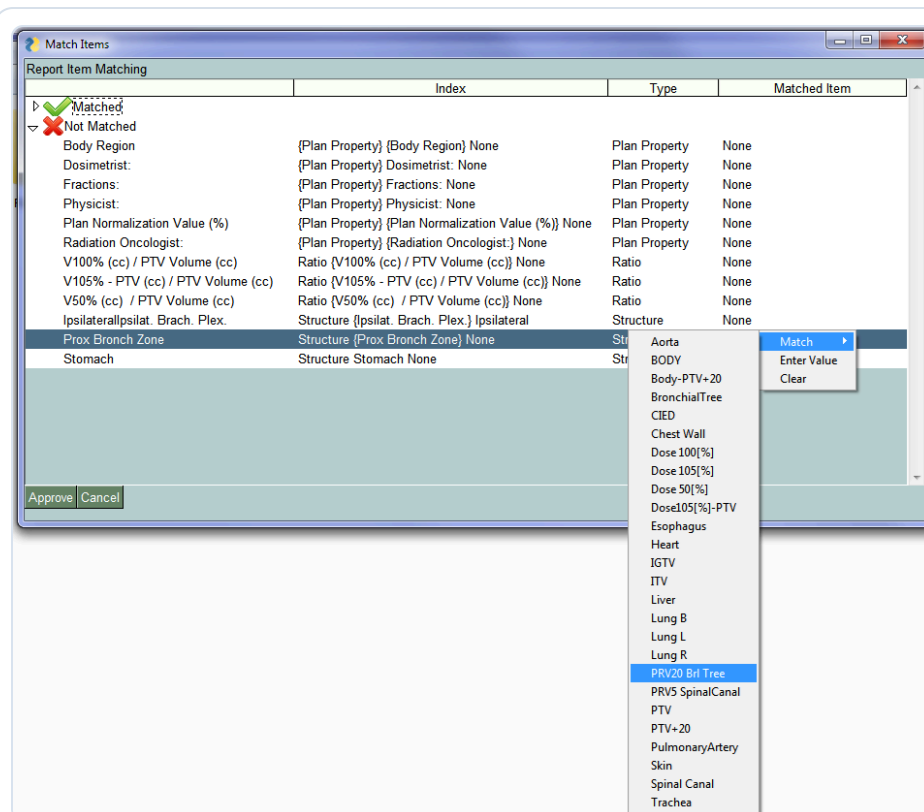
The items are grouped by whether the auto-match was successful or not.

Matching tries these steps in order for each report reference:

1. Direct name match to plan element.
2. Laterality-based name variation (using config lookup tables).
3. Alias lookup from global alias list and per-report aliases.

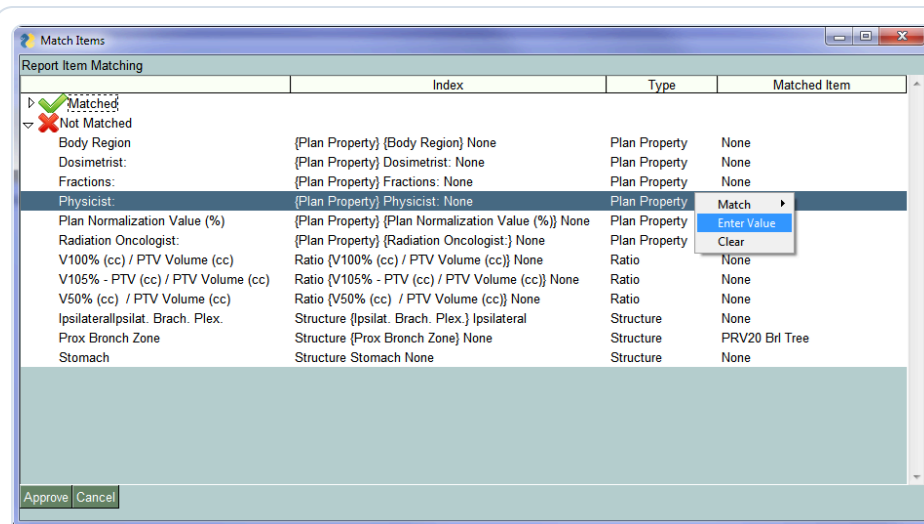
Manual match actions

- To manually match a structure to a given report item: right click on the desired item and hold the mouse over the Match heading on the menu that appears.
- Click on the appropriate structure from the list.

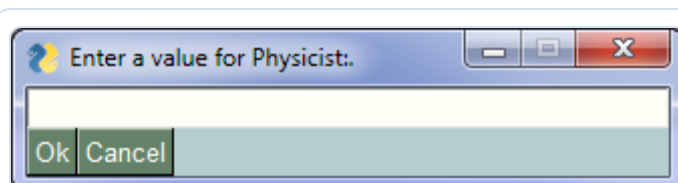


Step illustration: Manual structure selection from Match menu

- Selecting Enter Value from the right-click menu brings up a new window for typing in the text value.

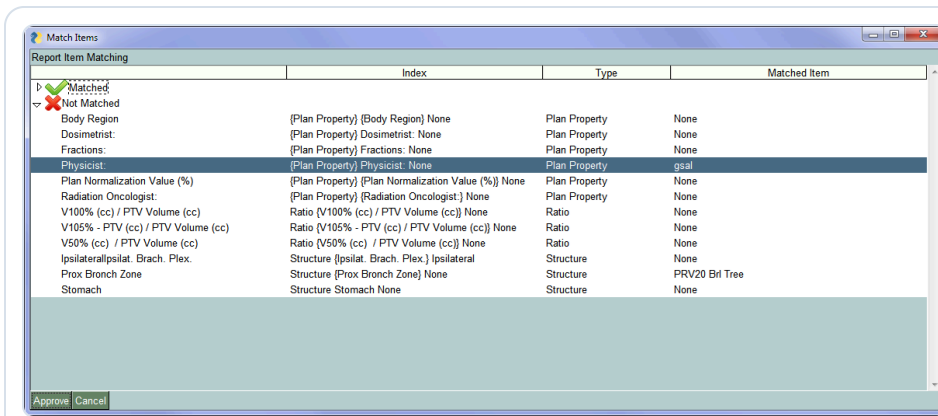


Step illustration: Choose Enter Value



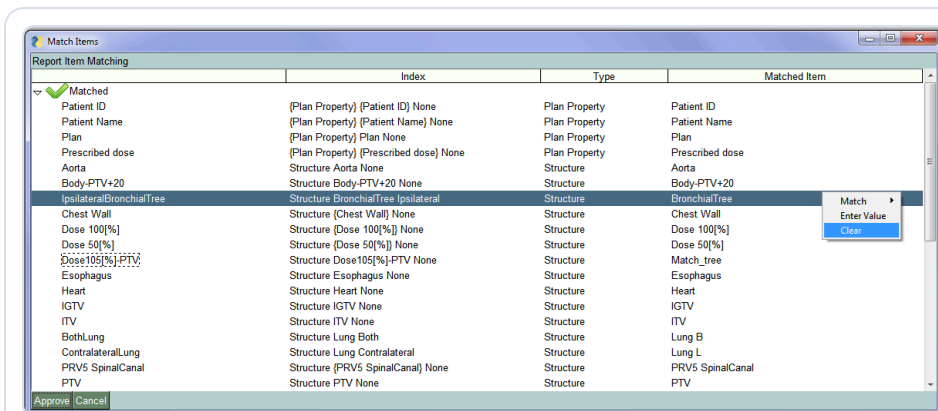
Step illustration: Enter a value dialog

- The value entered will display in the Matched Item column.



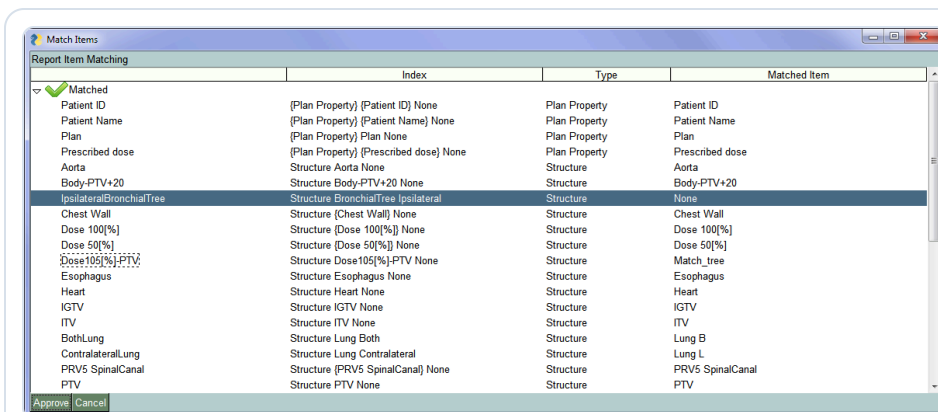
Step illustration: Entered value appears in Matched Item

- To remove a match, select Clear from the right-click menu.



Step illustration: Clear a match

- The value in the Matched Item column will become None.



Step illustration: Cleared item becomes None

- Once all desired modifications are complete, click on the Approve button in the lower left corner.

5) Generate And Review Report

1. Click Generate Report.

Plan Evaluation

Plan

LUNR **Plan Loaded**

Dose: 4800.0cGy
Course: C2
Exported February-14-2018 10:20:53

Patient:
Name: AA, BB
ID: 00000011

Report

SABR 48 in 4

SABR Plan Evaluation Sheet for 12Gy/fr Schedules (48 Gy in 4F) or (60Gy/5F)

Template:
File: SABR Plan Evaluation Worksheet BLANK.xlsx
WorkSheet: EvaluationSheet 48Gy4F 60Gy5F

SABR 48 in 4

Structured Matched

Generate Report

Exit

	Dose	Exported On
AA, B (00000011)		
LUNR	4800.0	February-14-2018 10:20:53
LUNL	4800.0	February-27-2018 10:04:16
LUNL	6000.0	March-05-2018 08:15:10
LUNL	6000.0	Friday, November 17, 2017 10:04:01

Step illustration: Structured Matched state, ready to generate report

2. The SABR Plan Report Form will appear and begin to populate.

	A	B	C	D	E	F	G	H	K	L
7			GTV Volume (cc)				7.40			
8			ITV Volume (cc)				7.40			
9			PTV Volume (cc)				24.00			
10			Total Lung Volume (cc)				4413.70			
11			Normalization: 100% Dose covers 95% of Target Volume							
12							Ref Dose (%) /	Calculated Value	Protocol	Acceptable
13							Ref Volume (cc)		Requirement	
14			Plan Normalization Value (%)				??		60%-95%	??
15			Dose @COM-PTV (cGy)				??		111.1%-166.7%	??
16			PTV Dose Inhomogeneity:							
17			PTV - Minimum Dose				90.8%		60%-95%	Yes
18			PTV - V100(%)				94.8%		95%	No
19			PTV - V90 (%)				100.0%		99%	Yes
20			HIGH Dose Spillage:							
21			Location V _{105%} - PTV (cc) =				0.00	0.00%	<15%	Yes
22			Volume V _{100%} (cc) =				25.50	1.063	<1.2	Yes
23			LOW Dose Spillage:							
24			Location D _{2cm} (%) =				61.5		54.7%	Minor Deviation
25			Volume V _{50%} (cc) =				105.90	4.41	4.47	Yes
26			DOSE TO OARs							
27							Maximum Dose (cGy) /	Calculated Value	Tolerance	
28							Volume (cc)		(cGy) / (cc)	
29			Lung Dose							
30			Mean Dose (contralateral lung)				62.4			??
31			Mean Dose (Total lung)				201.6			??
32			V20 (Total Lung)				V _{20Gy} (%) =		≤10%	Yes
33			Other OARs							
34			Aorta (max point dose)				979.2		6400	Yes
35			V60Gy ≤ 10cc V _{60Gy} =				0.00		10	Yes
36			Artery-Pulmonary (max point dose)				1046.4		6400	Yes
37			V60Gy ≤ 10cc V _{60Gy} =				0.00		10	Yes
38			Spinal Canal (max point dose)				600.0		3200	Yes
39			Ipsilat. Brach. Plex. (max point dose)				??		3800	??
40			V30Gy ≤ 3cc V _{30Gy} =				??		3	??
41			Heart (max point dose)				1425.6		6400	Yes
42			V60Gy ≤ 10cc V _{60Gy} =				0.00		10	Yes
43			Esophagus (max point dose)				1425.6		4000	Yes
44			Chestwall (rib) (max point dose)				5952.0		6000	Yes
45			Trachea Bronchus V50Gy ≤ 5cc V _{50Gy} =				4.45		5	Yes
46			Proximal Trachea (max point dose)				585.6		6400	Yes
47			& Bronch. Tree: V60Gy ≤ 5cc V _{60Gy} =				??		5	??
48			Stomach and Intestines V36Gy ≤ 1cc V _{36Gy} =				??		4000	??
49							??		1	??
50			Dosimetrist:							
51							??		??	??
52			Physicist:							
53							??		??	??
54										
55										

Step illustration: Generated report worksheet

3. Review Results.

43			Chestwall (rib)	(max point dose)	5952.0
44	Trachea	Bronchus	V50Gy ≤ 5cc	V _{50Gy} =	4.45
45	19.2	585.6	Proximal Trachea	(max point dose)	585.6
46	0.00	0.00	& Bronch. Tree: V60Gy ≤ 5cc	V _{60Gy} =	??

Step illustration: Trachea/Bronchus review detail

Note: values for both Trachea and Bronchus may appear, with the maximum used in the usual spot.

4. Clicking on any of the green buttons, or selecting a new dvh file or plan report, moves back to that step in the process.

6) Menus You Might Want to Use

Menu Action	Purpose
Select Plan DVH File	Open a specific DVH file directly when it does not appear in the list of available files.
Set Save File Name	Choose a folder and filename for the generated report.

7) Menu Items You Should Not Use

Menu Action	Purpose
Set Default Locations	Change DVH folder, report folders, template and save locations in config. This will change the locations for all users and may cause issues if set incorrectly. Contact a physicist if you need changes to these.
Update all Report Definitions	Rebuild report definition pickle from configured report folders. This is not needed for normal use and may cause issues if the report definitions are not correct.
Load Report Definition File	Load an additional report definition XML file. This is primarily intended for physicists to test new report definition XML files. If a new report definition XML file is needed, contact a physicist to create and test it.

8) Troubleshooting

Invalid Plan

- Cause: selected file is not valid DVH format or cannot be parsed.
- Action: verify extension and source export, then re-select file.

Missing Match Items

- Cause: naming mismatch between DVH structures and report references.
- Action: use manual matching, then update aliases for future automation.

Excel File Does Not Update

- Cause: output file is locked or workbook path is incorrect.
- Action: close conflicting workbook instances and verify save path.

9) Note

This program is a work in progress.

Some things will not work as intended.

Your feedback, bug reports, and feature requests are very welcome.

10) Clinical Use Caution

This tool supports plan evaluation workflow but does not replace independent clinical review. Verify output values in the worksheet before charting or sign-off.